

UNIVERSITY OF MALAKAND

**Patient Flow and Disease Trends: A Case Study of Medical
Department DHQ Hospital Timergara**



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ABSTRACT

This thesis presents a data-driven investigation of patient flow and disease trends and its future forecasting in the Medical Department of DHQ Hospital Timergara in order to understand and predict future trends. We investigated how patient traffic differs based on disease type, daily, weekly, monthly and demographics using records of patients admitted to the hospital.

It was observed from the patient flow study that admission trends in the Medical Department of DHQ hospital Timergara were quite predictable with seasonal variability. Number of patients were higher in summer and autumn with a number of consistent spikes occurring annually at defined times of the year. This study highlights the role of seasonable predictions and forward planning.

Analysis of the disease indicated that elders were carrying the highest burden of disease with Diabetes Mellitus, Hypertension and Stroke being the primary causes of admission among the chronic illnesses and Acute Gastroenteritis and Lower Respiratory Tract Infection being prevalent for the seasonal illnesses showing significant variations in their presentation in summer and winter. It was evident that there is need for continuing chronic disease management while trying to manage predictable rise of communicable diseases seasonally.

For all the diseases analyzed, Hypertension is with 8.23% MAPE had the lowest errors, followed closely by Lower Respiratory Tract Infection with 10.64% MAPE. Both error values are considerably low and represent high confidence and stability of forecasts generated by the model, suggesting that the pattern of these diseases over time could be very consistently predicted. MAPE for Tuberculosis is 19.55%, Fits is 25.93%, Diabetes Mellitus is 30.38%, Cerebrovascular Accident is 30.62%, Organophosphorus Poisoning is 31.84% and Pneumonia is 37.36% had similarly low to moderate forecasting errors, suggesting a relatively consistent trend and therefore easier and more reliable prediction.

Medium performance predictions of diseases over time were found. MAPE for Sepsis is 47.73%, Blood Transfusion is 51.47%, Chronic Obstructive Pulmonary Disease is 53.72%, Depressive Illness is 55.15%, Poisoning is 56.94% is and Diabetic Ketoacidosis is 79.13%. Though higher than for earlier mentioned diseases, the models were still effective at charting and interpreting trends, even if slightly more errors were being made during each prediction.

Larger forecasting errors were made, MAPE for Headache is 107.43%, Vertigo is 111.78%, Shortness of Breath is 114.48%, Aspiration is 124.31%, Mushroom Poisoning is 144.74%, Cough is 146.76% and Snake Bite is 157.10%. These highly varying predictions can be attributed to significant and rapid fluctuations in the trends and prevalence of these diseases. Possible reasons could include the possibility of outbreak, weather, and seasonality influencing the figures, or simply due to randomness in patient admission of such ailments.

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CHAPTER 1

INTRODUCTION

Every day, large volumes of patient data are generated by hospitals from admission up to the diagnosis phase. Analyzing these data helps them improve healthcare delivery, managing patient flows within the facility and identify community diseases' trends. Proper utilization of such data becomes therefore indispensable to achieve better health services. Data analytics application in the healthcare sector in recent years continues to grow because of its role to support evidence based decision making and enhancing hospital management.

Patient flow means movement of patients through all the process stages in a hospital from admission until their discharge from the hospital. Effective management of patients' flows contributes to avoiding overcrowding, reducing waiting times, improving allocation of resources and consequently leading to enhanced quality of healthcare services. Disease trends analysis, also helps health providers to better understand the prevalence and distribution of diseases in terms of time, population groups of different age, season etc.

Many hospitals in developing countries are still utilizing paper based record keeping systems and have patient's data in registers. Those records though include important health related information are mostly underutilized since no systematical approach is available for data analysis. Through digitalization of patient's records in hospitals and analytical techniques, important trends could be clearly revealed and utilized to better inform health care decision and planning.

DHQ Hospital Timergara is a primary health care provider to the region of Lower Dir and surrounding region of Khyber Pakhtunkhwa Pakistan. The hospital is facing variable patient load and different number of cases all year round but no research has been conducted to study the trends of patients' flows and diseases cases at the medical department of DHQ Hospital Timergara.

Development of data analysis and visualization tools like, programming languages and visualization software make possible to analyze hospital data in better way. Python language is used with a number of libraries available that ease data preprocessing, exploratory data analysis, trends analysis and forecasting. Power BI can create interactive visualization that facilitates presentation of large volume of complex healthcare data to technical and non-technical people.

In this research, patient data of medical department of DHQ Hospital Timergara is collected from hospital registers and converted into Microsoft Excel format and analyzed with Python programming language. Analysis focuses on the most common diseases cases and their frequencies of occurrence, patterns of patient's admissions over time, variation of diseases' cases according to the age and season and forecasting the number of patients' cases at the medical department in the future using historical data.

The findings from this study are expected to help hospital management optimize their resource planning, patient management and health care delivery system via providing comprehensive insight from the data analysis. Furthermore, this research aims to give a case of effective use of data analysis and visualization tools in the public sector hospitals.

1.1 Define Patient Flow and Disease Trend

Patient flow relates to the movement of patients through the various stages of care within the hospital from admission to discharge, while disease trends relate to the occurrences and variations in diseases over time in a population.

1.2 Background of the Study

The health system generates vast amount of data in form of patient admissions, their diagnoses and treatments, which is underutilized or un-analysed in the management systems of many public sector hospitals. Thus hospital management usually works on assumptions instead of facts in planning of its resources, staff and medical service delivery.

There are many a challenge with the regional health organizations like DHQ hospital Timergara, including fluctuating patient admission, resource limitations, structured data analysis limitation, etc. Information about the patient movement within the hospital, variation in diseases over seasons and age groups are necessary to provide a better healthcare and efficient functioning of the hospital.

In the field of data science and data analytics, useful information could be extracted from data pertaining to old patient records within the hospital. Data pre-processing, Exploratory data analysis and visualization could uncover interesting patterns of patient flow and disease prevalence. Such insights can lead to better healthcare decision making.

In this project, hospital register data is digitized into MS Excel and analysis using Python program and Power BI visualization tool, and to investigate patient flow and trends in medical department of DHQ hospital Timergara.

1.3 Problem Statement

Although records of the patients are available, there is no systematic study and analysis on the trend and patient flow of the diseases at DHQ Hospital Timergara. Lack of informed information has led to inadequate use of resources, inefficient preparation for the patient workload and less knowledge of the trend of disease at various segments of population.

Most of the records of patients are manually kept in registers that help very little to derive trend and provide meaningful reports useful to decide. It may be challenging for the hospital management to estimate its resources, staffing level, and disease trends.

So the purpose of this research is to make use of old patient's record to analyze the trend and patient flow within the medical ward of DHQ Hospital Timergara using health analytics tool.

1.4 Objectives of the Study

Objectives of this study are:

- To analyze patient flow over time in the medical department
- To identify common diseases and their distribution
- To examine disease patterns across different age groups
- To analyze seasonal variations in disease occurrence
- To visualize trends for better interpretation and reporting
- To explore basic forecasting techniques for predicting future patient load and disease trends

1.5 Research Questions

Questions to be answered in this study:

- Which diseases are most frequent among the patient cohort?
- How does the number of admitted patients change over time?
- Which are the most sensitive age groups for particular diseases?
- Which are the diseases that are growing or decreasing over time?
- Are there any particular seasonal tendencies of diseases?
- Can future patient numbers and disease trends be predicted using historical data?

1.6 Significance of the Study

The significance of this study lies in its applicability to both technical and non-technical decision makers. For the hospital management, this provides statistical evidence that will facilitate resource planning, bed occupancy management and enhancement of service quality.

It allows non-technical decision makers access to simplified information, which is derived from complex data, through easy to understand visualizations and aggregate trends. This would be of great interest to researchers and students, illustrating a potential real-world application for data analysis skills.

To the field of healthcare analytics, it is a case study providing evidence from a local public hospital and thus research within a regional context in Pakistan. The utilization of Python and Power BI also exemplifies contemporary analytical tools and applications.

1.7 Scope of the Study

Patient data are gathered from the medical department of DHQ hospital Timergara. They are obtained from hospital registers and then converted into Microsoft excel format for the analysis.

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Trend analysis

- Basic forecasting techniques

Power BI is used for:

- Data visualization
- Dashboard creation
- Trend reporting

The study mainly examines:

- Patient admissions
- Disease occurrence
- Age-group analysis
- Seasonal disease patterns
- Forecasted patient trends

1.8 Limitations of the Study

In this study, data is collected from the patient's records, under the category of medical department of DHQ Hospital Timergara. It is obtained from the registers of hospital and then transformed in the Microsoft Excel.

The study has following limitations:

The study is based on recorded historic data, which is likely to contain missing values or inconsistent data.

The study conducted only in one department; hence, the conclusion drawn may not apply to the whole hospitals.

Some records are missing in their relevant data because it is recorded manually.

Forecasting methods used in this study are of explorative type and so the predictions may not be accurate to high level of confidence.

No factors like environmental, economic or social situations have been accounted for in the analysis.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter deals with existing literature on patient flow analysis, disease trends analysis, health care data analytics, seasonal diseases pattern and forecast at hospital level.

This literature review is necessary to set the theoretical and empirical background for the study of "Patient Flow and Disease Trends: A Case study of Medical Department DHQ Hospital Timergara". Concepts of patient flow and disease trends, related literature already published and justification for present study is addressed in this chapter.

Research shows that the healthcare data can play a crucial role in patient management, disease trends and for decision-making in hospitals. It also highlighted the role of admission forecasting, and disease seasonal trends for better planning of resources in the hospital.

2.2 Literature Review

2.2.1 Patient Flow in Hospitals

Patient flow represents the transition of patients between different health care stages, that is registration, diagnostic, therapeutic, admission, transfers, discharge. Efficient patient flow has a direct impact on waiting time, patient satisfaction and hospital resources optimization. As mentioned by Palmer et al. (2018), patient flow management in hospitals plays an important role in their effectiveness as congestion and delays could harm the healthcare service. Effectively managed patient flow enables healthcare service provision in a timely and orderly manner. There are numerous challenges facing hospitals, including patient increasing demand, limited staffing resources and insufficient infrastructure, resulting in congestion of patient flow particularly in public hospitals. El-Bouri et al. (2021) illustrated that data analysis and forecasting techniques are being increasingly applied by health care organizations to observe and forecast patient flow and future admission demand.

Analysis of patient flow commonly involves:

- Daily and monthly patient admissions
- Bed occupancy
- Waiting times
- Admission and discharge rates
- Patient demographics
- Disease-specific patient movement

In this study, patient flow focuses mainly on patient flow and disease distribution within the medical department of DHQ Hospital Timergara.

2.2.2 Disease Trends

Disease trends refers to any trend or change in the occurrence of a disease in a population over a period of time. With disease trend analysis health workers get a broad understanding on how a disease occurs and also whether its occurrence is decreasing or increasing over a specific period of time.

Disease trends could vary because of:

- Seasonal factors
- Environmental conditions
- Population demographics
- Lifestyle changes
- Epidemics and outbreaks

Simpson et al. (2022) note that diseases, particularly infectious diseases are likely to change based on the seasons and using trend analysis would help hospitals recognize and plan for periods of higher patient demand.

Disease trend analysis in healthcare institutions used for:

- Early disease detection
- Resource planning
- Public health surveillance
- Forecasting outbreaks
- Making better health care policies

The present study examines disease trends among patients admitted to the medical department of DHQ Hospital Timergara and identifies patterns across age groups and seasons.

2.2.3 Healthcare Data Analytics

Health Care Data Analysis is the use of statistical and computational techniques to derive useful insights from health care data. Hospitals produce huge amounts of data such as EMRs, lab reports, admission record and diagnosis history.

Health Care Data Analysis comprises primarily:

- Data collection
- Data cleansing and preprocessing
- Exploratory Data Analysis (EDA)
- Visualization
- Trend analysis
- Forecasting and prediction

According to Jiang et al. (2022), healthcare analytics allows hospitals to discover non-obvious trends that might exist within the patient admission data and disease occurrences. Information on healthcare patterns can then be interpreted by stakeholders via use of charts and descriptive statistics.

Tools that are commonly used:

- Python
- R programming
- Tableau
- Power BI
- Excel
- SQL databases

2.2.4 Seasonal Disease Patterns

Many diseases show seasonal behavior. Respiratory infections, flu, malaria, dengue fever and gastrointestinal diseases often increase during specific seasons. Seasonality refers to repeated patterns occurring during certain periods of the year. According to Silva et al. (2023) seasonal analysis helps hospitals anticipate patient admissions and prepare healthcare resources accordingly.

Seasonal disease analysis assists in:

- Staffing decisions
- Medicine stock management
- Bed allocation
- Emergency preparedness

The current study investigates whether diseases in DHQ Hospital Timergara exhibit seasonal variations over time.

2.2.5 Forecasting in Healthcare

Forecasting involves predicting future events based on historical data. In healthcare, forecasting is used to estimate future patient admissions, disease outbreaks and hospital workload.

Common forecasting techniques include:

- Moving averages
- Linear regression
- Time series analysis
- ARIMA models
- Seasonal ARIMA (SARIMA)

As it was observed by Gul and Celik (2020), statistical forecasting approaches offer hospitals great opportunities for improvement in the context of planning and patient management. Using forecasts, hospitals can better predict future patient volumes and manage their resources.

2.3 Theoretical Framework

2.3.1 Systems Theory

The theoretical framework of this study is Systems Theory. The hospital is a system, and its elements are mutually interacting departments or functions. Systems Theory, and in a healthcare system changing one part will make other part changing too.

The patient flow healthcare process as below:

- Inputs = patient admissions
- Processes = diagnosis and treatment
- Outputs = discharge and healthcare outcomes

Efficient management of patient flow improves the overall performance of the hospital system.

2.3.2 Time Series Theory

Time Series Theory is used to describe data recorded over a period of time and to examine patterns such as trend, seasonality, cycles and erratic variation. It has extensively been applied in healthcare forecasting to predict future number of admissions, incidences of disease, and future hospital resource requirement by using the existing data over a period of time. In Pakistan, the application of Time Series Theory on a larger scale was conducted by a study at Mayo Hospital Lahore using the model ARIMA to forecast number of patients entering the Outpatient Medical Laboratory (OPML). This study used patient data ranging from 2009 to 2013 and found that ARIMA was able to successfully predict number of patients entering. This shows the reliability of the forecasting methods in hospital management and resource planning. (The Professional Medical Journal). Recently, researchers in healthcare forecasting are using the more advance time series forecasting methods such as Prophet for predicting hospital bed occupancy and patient movement at hospital, the Prophet model separates out the trend and seasonality component of historical data allowing health workers to better anticipate future demand and manage their resources efficiently. Hence, time series theory is the basis of the current study, where patient admission trend and disease trend data over a period of time was extracted from DHQ hospital Timergara in order to develop future disease incidence forecast by using the Prophet forecasting model.

2.4 Empirical Literature Review

2.4.1 Studies on Patient Flow

Betrsimas et al. (2002) Predicted inpatient flow using interpretable analytics. They revealed that patient admission forecasting enhances hospital capacity planning and decrease crowding in hospitals. El-Bouri et al. (2021) Surveyed applications of machine learning to patient flow management, concluding that predictive models could benefit hospital operations. Palmer et

al. (2018) Evaluated operational research models in patient flow modeling, arguing that health care analytics aids in decision making and resource management. Susnjak and Maddigan (2023) researched forecasting of patient flow during epidemics, suggesting that historical trends and seasonal patterns help predict patient admissions.

2.4.2 Studies on Disease Trends

Using time series models, Simpson et al. (2022) explored the seasonality of infectious diseases and revealed strong seasonal patterns in diseases occurrence. It suggests the need for time-series analyses for monitoring of diseases. In order to address the challenge of predicting the arrival of patients and the analysis of patients flow in hospitals during COVID-19, time series models were used by Tavakoli and Tavakkoli-Moghaddam (2022) and the forecasting models can be helpful for hospitals when dealing with a health crisis. Cao and Zhang (2025) studied outpatient visits trend by the SARIMA model and concluded that time series forecasting can effectively enhance healthcare resource allocation and planning.

2.4.3 Studies on Healthcare Analytics

Jiang et al. (2022) conducted a systematic review of patient arrival modeling and found that data analytics and forecasting significantly improve healthcare operations. Miezah (2025) examined predictive analytics for healthcare resource optimization and concluded that healthcare institutions can improve efficiency through data-driven planning. Atobatele et al. (2019) studied big data analytics in population health management and highlighted the role of predictive modeling in disease prevention and healthcare optimization.

2.5 Research Gap

Previous studies have extensively explored patient flow forecasting, disease trend analysis and healthcare analytics in developed healthcare systems. However, limited research has been conducted in regional public hospitals in Pakistan particularly in DHQ Hospital Timergara.

Most previous studies showed:

- Advanced machine learning models
- Emergency departments
- Large-scale healthcare systems

Very few studies have analyzed:

- Disease distribution across age groups in regional hospitals
- Seasonal disease patterns in local healthcare settings
- Patient flow trends in DHQ hospitals
- Practical exploratory analytics for healthcare decision-making in Pakistan

Therefore, this study addresses this gap by analyzing patient flow and disease trends using historical hospital data from the medical department of DHQ Hospital Timergara.

2.6 Conceptual Framework

The conceptual framework of the study is based on the relationship between hospital patient data and healthcare decision-making.

2.6.1 Independent Variables

- Patient demographics
- Admission dates
- Disease diagnoses
- Seasonal factors

2.6.2 Analytical Processes

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Trend analysis
- Visualization
- Forecasting

2.6.3 Dependent Variables

- Patient flow patterns
- Disease trends
- Seasonal disease behavior
- Forecasted patient load

2.7 Summary

In this chapter, I have reviewed studies relating to patient flow, diseases patterns, health care analytics, seasonal health patterns and forecasting models. As from the above review, health care data analytics provides a wide spectrum of ways to optimize hospital management and healthcare planning, trend analysis and forecasting studies were proved to be useful to estimate hospital patient admission and disease incidence.

The chapter also identified a research gap regarding limited studies on patient flow and disease trends in regional hospitals of Pakistan. The present study seeks to fill this gap by analyzing patient flow and disease trends in DHQ Hospital Timergara using exploratory data analysis and basic forecasting techniques.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains the methodology used in the study titled “Patient Flow and Disease Trends: A Case Study of Medical Department DHQ Hospital Timergara.” It describes the research design, study area, data sources and data collection procedures, data preprocessing methods, analytical techniques, software tools and ethical considerations used during the research.

The methodology is designed to analyze patient flow and disease trends using hospital records converted into digital format for statistical analysis and visualization.

3.2 Research Design

This study uses a **quantitative research design** based on exploratory and descriptive data analysis techniques. The research will employ a quantitative method because the research work is based on numerical hospital data such as number of patients admitted, different age groups, record of disease and time of admission. The research also follows a case study on medical department of DHQ hospital Timergara for in-depth study on patient flow and trend of disease within the healthcare set up.

The research will involve:

Exploratory Data Analysis (EDA)

- Trend analysis
- Descriptive statistics
- Visualization techniques
- Basic forecasting methods

These methods help identify patterns, relationships, and seasonal changes in hospital patient data.

3.3 Study Area

This study is carried out on Medical department of District Headquarters (DHQ) hospital Timergara which is located in lower Dir district Khyber Pakhtunkhwa Pakistan. The DHQ hospital Timergara serves a large population from nearby both urban and rural areas and has a range of services available, this is covered by different departments which include medical department, emergency unit, surgical unit, outpatient department and so on. The medical department was chosen due to the large volume of admitted patients and disease prevalence which makes it an excellent department for observing flow of patients and diseases.

3.4 Data Sources

The data was collected from the registers of the patients from the medical department in the DHQ Hospital Timergara.

The collected data was made up of the following attributes:

- Patient Name
- Age
- Gender
- Address
- Diagnosis/disease
- Admission date
- Discharge date
- Department information

3.5 Data Collection Procedure

The data collection consists of following steps:

Step 1: Permission and Approval

Patient admission forms were accessed from hospital records after receiving consent from the hospital administration.

Step 2: Data Extraction

Information was collected from the hospital registers managed by the medical department, and information was manually abstracted from it.

Step 3: Data Digitization

The collected register data were put into Microsoft Excel files. All patient records were set in rows and columns.

Step 4: Data Verification

Care was taken while looking through the entered data to recognize errors in typing, extra entries and omission of any kind of data.

3.6 Data Preparation and Preprocessing

Raw healthcare data sometimes includes anomalies such as inconsistencies, empty values, formatting differences and others. Hence, before analyses, preprocessing steps were taken.

The preprocessing process consist of following steps:

3.6.1 Data Cleaning

Data cleaning involved:

- Removing duplicate records
- Fix spelling variation
- Standardizing disease names

- Handling missing values
- Removing incomplete entries where necessary

3.6.2 Data Transformation

To prepare dataset for analysis data transformation is done.

Following transformation is performed:

- Converting dates into proper date time format
- Extracting day, weeks and months from admission dates
- Categorizing patients into age groups
- Creating disease categories

Example age groups:

- Children (0–12 years)
- Teens (13-19 years)
- Young-adult (20–35 years)
- Adults (36–50 years)
- Senior (51-65 years)
- Old (66+ years)

3.6.3 Handling Missing Values

Missing values were managed through:

- Removal of highly incomplete records
- Filling minor missing values where possible
- Using “Unknown” category for some missing disease records

Proper handling of missing values improves analysis accuracy.

3.7 Tools and Technologies Used

The study uses Microsoft Excel, Python programming language and Power BI for data analysis and visualization.

3.7.1 Microsoft Excel

Microsoft Excel was used for:

- Initial data entry
- Data storage
- Organizing patient records
- Preliminary filtering and sorting

Excel is used for converting hospital register data into digital format.

3.7.2 Python Programming Language

Python was used for:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Statistical analysis
- Trend analysis
- Forecasting

For healthcare analytics the programming language (Python) and their libraries.

3.7.3 Python Libraries Used

Some of the python libraries that have been employed at various stages of the data processing, analysis, visualization, and forecasting are mentioned below:

1. Pandas

This library has been used for data handling, manipulation, cleaning, transformation and preprocessing the data acquired from hospitals. Numpy- Numpy library had been extensively used for mathematical and numerical operations as well as handling multidimensional arrays.

2. Matplotlib

This library was used for plot generation, chart creations and analyzing the trend.

3. Seaborn

A sophisticated graphical representation of data that increases clarity.

4. Scikit-learn

The machine learning library was used for some initial predictive analysis and validation of models used for forecasting.

5. Prophet

This library was utilized for time series forecasting, in order to be able to appropriately capture trend and seasonality, particularly in hospital data. The library helped to build a strong prediction framework for disease prevalence for future time periods based on previous trends.

3.7.3 Power BI

For interactive visualization and dashboard, we used Power BI.

Power BI helped generate:

- Admission trend charts
- Disease distribution graphs
- Seasonal trend visualizations
- Age-group comparisons
- Forecasting dashboards

3.8 Data Analysis Techniques

3.8.1 Exploratory Data Analysis (EDA)

To understand the dataset, we performed EDA.

EDA techniques included:

- Frequency distributions
- Descriptive statistics
- Correlation analysis
- Trend visualization

EDA helps identify:

- Most common diseases
- Peak admission periods
- Age-group disease distribution
- Seasonal variations

3.8.2 Descriptive Statistical Analysis

To summarize the data numerically descriptive statistics were used.

The following measures were used:

- Mean
- Median
- Frequency counts
- Percentages
- Standard deviation

To understand demographics and disease occurrence these measures was used.

3.8.3 Trend Analysis

To understand the patient flow and disease occurrences over time the trend analysis is used.

Focused of analysis:

- Monthly admission trends
- Yearly disease patterns
- Increasing or decreasing disease occurrence

To identify patterns Line charts and time-series graphs were used.

3.8.4 Seasonal Analysis

To determine diseases, occur more frequently during specific seasons or months seasonal analysis was performed.

The study shows:

- Summer disease patterns
- Winter disease patterns
- Seasonal patient admission changes.

3.8.5 Forecasting Techniques

To estimate future patient admissions and disease trends basic forecasting techniques were used.

The forecasting methods included:

- Moving averages
- Time-series forecasting
- Linear trend analysis

Forecasting provides insights into possible future healthcare demand.

3.8.6 Output or Forecast of the Prophet Model

In a time-series forecasting thesis (particularly when utilizing algorithms like Facebook Prophet), the terms `ds`, `yhat`, `yhat_lower`, and `yhat_upper` represent the core output structure of your predictive model.

- `ds` (Datestamp / Time-axis)

The independent chronological variable representing the specific time execution interval of the model.

`ds` stand for "datestamp." This is the base time axis for past data and the prediction horizon into the future. This would correspond to days, weeks, or months depending on the resolution you are looking at (e.g., 2024-07-01). It is used to determine how a model can isolate and model time series variations (e.g., seasonal increases in gastrointestinal illness in the summer, or Holiday decreases).

- `Yhat` (Predicted Point Forecast)

The estimated conditional mean (or optimal point estimate) calculated by the forecasting model at any specific point in time (t).

The point forecast, `yhat`, is the main output from the predictive model, indicating the expected number of patient cases or admissions on a specific date. It is considered the algorithm's "best guess" or central tendency of what will occur after estimating the basic trend and adding the seasonal influences (year/week) as well as any special holidays.

Impact to Hospital Operations: Yhat is used by hospital administrators as a base for normal operations such as scheduling normal nurse work shifts, ordering normal amounts of pharmaceutical drugs, and expecting a base amount of bed capacity to be occupied.

- Yhat_lower & yhat_upper (Uncertainty Intervals)

The predicted lower and upper bounds on the uncertainty in the forecast made by the model. This is usually defined using the (95%) confidence intervals.

Flow of patients in reality is rarely exactly linear but rather fluctuates and it is convenient to have a "statistical safety net" of uncertainty around a point prediction (yhat).

yhat_lower represents the minimum expected patient volume.

yhat_upper represents the maximum expected patient volume.

3.9 Visualization Techniques

Tools for visualization.

The study uses:

- Bar charts
- Line graphs
- Pie charts
- Heat maps
- Time-series plots
- Dashboard reports

For visualization Python and Power BI both were used.

3.10 Ethical Considerations

Throughout the study ethical considerations is followed.

These measures were taken:

- Protection of patient confidentiality
- From analysis personal identifier were excluded
- For academic purposes this data was used only
- Hospital authorities gives the permission

No sensitive personal information was disclosed in the research.

3.11 Limitations of Methodology

Several limitations were present in the study:

- The data depends on manual hospital records which may contain errors
- There are several records may have missing or incomplete information
- The study focuses on one department of hospital only
- Forecasting methods are basic and exploratory

Despite these limitations, the methodology provides useful insights into patient flow and disease trends.

3.12 Summary

In this chapter the research methodology adopted to conduct the research is discussed. A quantitative case study research design has been adopted in order to identify the flow of patients and to assess the trends of diseases by using the secondary data provided by the Medical Department of DHQ Hospital Timergara. The data has been collected from hospital registers and then converted into a formal Microsoft Excel dataset to carry out various analyses. The study has used Python programming language to extract, cleanse, statistically analyze and predict the data, while Power BI has been used for data visualization.

The methodology followed by the study comprises the following stages:

Data collection, Data cleansing, Exploratory Data Analysis, Trend analysis, Seasonal pattern analysis and to forecast by applying a simple forecasting model. It also helped in recognizing various patterns and trends in the data, including anomalies and variations over the time. Ethics have also been taken into consideration.

CHAPTER 4

RESULTS

4.1 Introduction

In this chapter, analysis and interpretation of data, collected for study on "Patient flow and disease pattern: A case study of medical department, DHQ Hospital Timergara" will be carried out. Patient demographic profile, patient flow trend, disease trend, admission/discharge status, future forecasting trend and dashboard visualization were observed. The hospital data collected was analyzed using descriptive statistics such as frequencies, percentages, trend, forecasting, table and graphic analysis and interpretation.

Some inferences regarding trends in healthcare utilization, disease burden, seasonality and future demand for health services are available through the finding.

4.2 Interpretation of Patient Flow Findings

This part contains interpretation of patient flow analysis results that was taken from the examination of patient's records of the medical department of the DHQ hospital Timergara. Various graphs and charts were presented to highlight on different issues of patient's data such as different type of patients, admission rates, success of treatments etc. All the graph had been analyzed and major features had been identified that could help management to identify and manage the patient flow patterns.

4.2.1 Patient Flow Analysis Over Time (Daily)

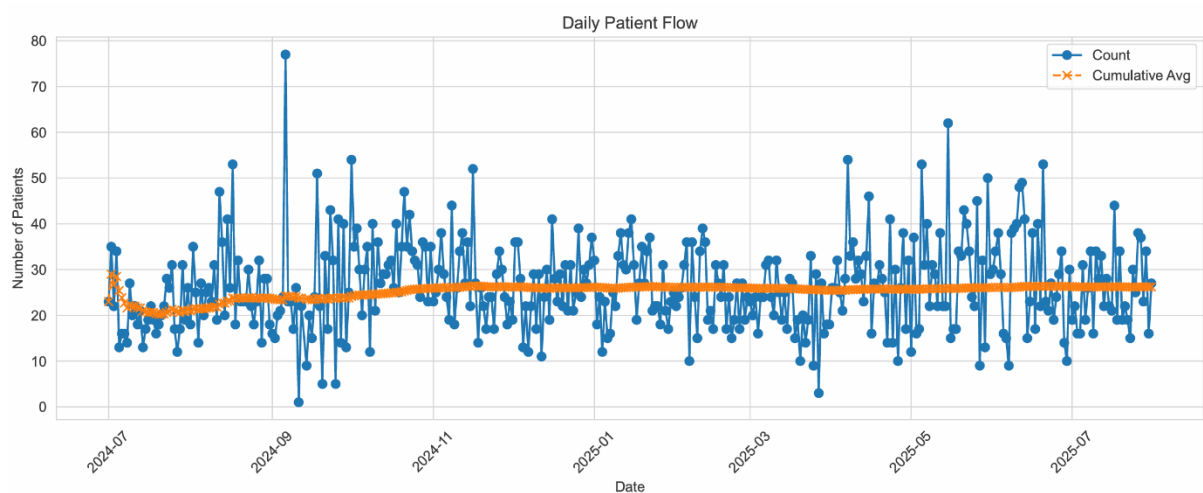


Figure 4.1: Daily Patient flow.

4.2.2 Patient Flow Analysis Over Time (Weekly)

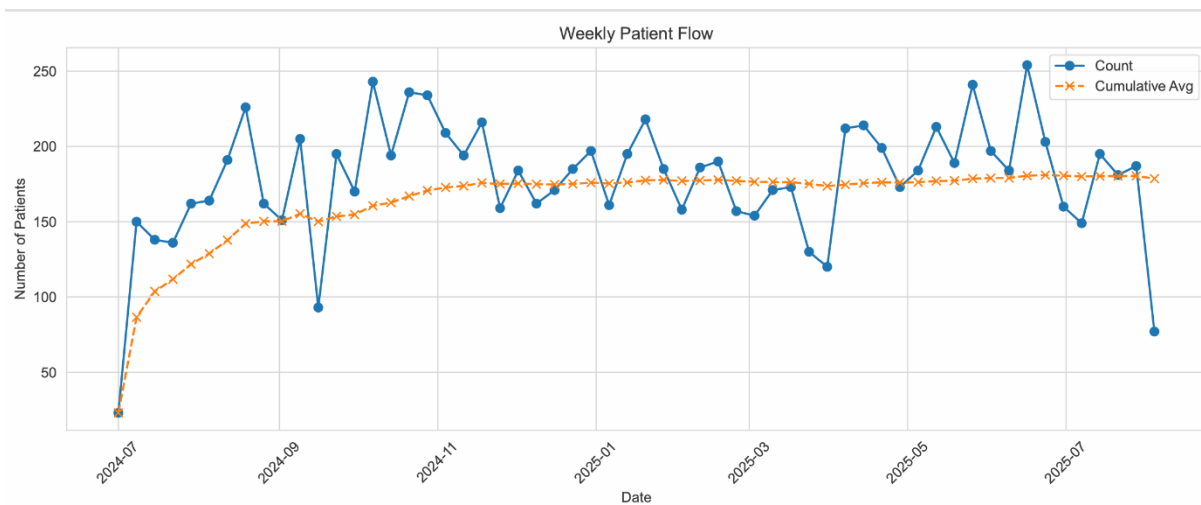


Figure 4.2: Weekly Patient Flow

4.2.3 Analysis Over Time (Monthly) Patient Flow

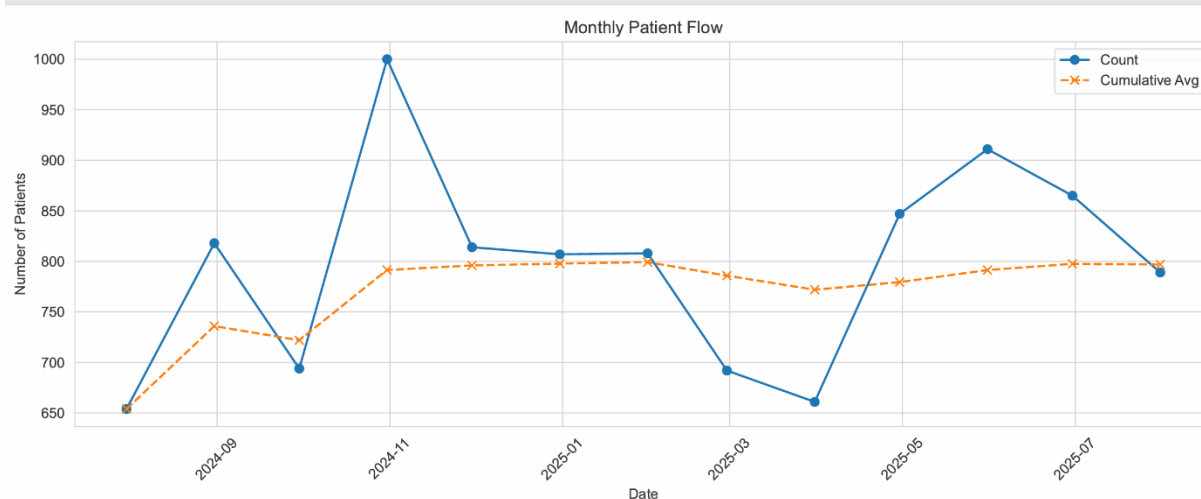


Figure 4.3: Monthly Patient Flow

From the interpretation of above figures 4.1, 4.2 and 4.3 Daily, Weekly and Monthly Patient Flow the following insight are collected.

Daily Patient flow: The figure 4.1 illustrate that during the study period the maximum patient flow in the medical department is observed in September 2024 which is reached an absolute maximum of 77 patients in a single day.

While the minimum patient flow is also in September 2024 of 1 patient in a day.

The average arrival of patient represented by the orange (x) line. Cumulative average line represents a range of 22 to 27 patient admission per day.

Weekly Patient flow: The figure 4.2 illustrate that maximum weekly patient flow reached to 254 patients in the second week of June, 2025.

While the minimum patient in a week during study period is 23 patients in the first week of July,2024. Average weekly patient flow is represented by (x) orange line which is ranging from 170 to 180 patients.

Monthly Patient flow: The figure 4.3 illustrate that maximum monthly patient flow reached to 1000 patients in the month of October,2024. While the minimum patient in a month during study period is 654 patients in the month of July,2024. Average monthly patient flow is represented by (x) orange line which is ranging from 780 to 796 patients.

4.2.4 Patient Flow Seasonality

From the analysis of patient data (July,2024 to July,2025) of medical department DHQ hospital Timergara we identify the seasonality of patient flow.

During the season of Summer, the patient flow is maximum (3126) and Autumn season also show high number patients (2508) compare to other seasons (Spring and winter).

While the Spring and winter seasons show the similar behavior of patient flow (2419) and (2307) respectively.

4.2.5 Spike Detection in Patient Flow

During the study of Medical department dataset, a total of **15 statistical spikes** were detected (where patient traffic significantly exceeded normal variance).

Date	Number of Patients	Percentage
September 6, 2024	77	194% of average / +50.8 patients
May 15, 2025	62	136% of average / +35.8 patients
October 1, 2024	54	106% of average / +27.8 patients
April 7, 2025	54	106% of average / +27.8 patients
August 17, 2024	53	102% of average / +26.8 patients
May 5, 2025	53	102% of average / +26.8 patients
June 20, 2025	53	102% of average / +26.8 patients
November 16, 2024	52	98% of average / +25.8 patients
September 18, 2024	51	94% of average / +24.8 patients
May 30, 2025	50	91% of average / +23.8 patients
June 12, 2025	49	87% of average / +22.8 patients
June 11, 2025	48	83% of average / +21.8 patients

Date	Number of Patients	Percentage
August 12, 2024	47	79% of average / +20.8 patients
October 21, 2024	47	79% of average / +20.8 patients
April 15, 2025	46	75% of average / +19.8 patients

The analysis of spike detection in the data shows that hospitals experiences high predictable fluctuations. Patient traffic is not arbitrary, operational noise, but follows significant patterns in Late Summer/Early Autumn (Aug-Oct) and Late Spring/Early Summer (Apr-June). Given this structural reality, hospital administrators need to shift focus from day-of emergency response to seasonal proactive staffing and resource planning during these critical block of the year.

4.3.6 Discharge Type Analysis

Discharge type analysis give the idea about the outcomes of the department treatment. So, from the analysis of discharge type in Medical department we identify that the stable and dominance discharge type is DOR or routine discharge with 82.12%.

While the cases of (REFERRED) and (LAMA) is roughly balanced at 7.0% and 6.5% respectively.

DEATH rate is consistent at 4.37% on average throughout the duration of the study.

4.3.7 Length of Stay

Length of stay shows the time taken by a patient in medical department.

- So, 82.12% of patients associated with discharge type DOR is average length of stay is (1.30 days).
- In the study it is identified that the maximum average LOS in medical department of DHQ hospital Timergara is associate with DEATH average (1.53 days) which 4.37% of total discharge during the study period.
- While LAMA 6.53% of the total patient population with the shortest length of stay of 1.13 days.

4.3 Insights of Disease Trends

This section presents the interpretation of disease trends findings obtained from the analysis of patient records in the Medical Department of DHQ Hospital Timergara. The result of those finding are discussed below.

4.3.1 Disease Analysis by Group Age

From the analysis of disease over different group of age the following insight are taken.

- The most affected age is Old (66+ years) 30.76% about the total cases of 2653.

- Next affected age is Senior (51-66 years) 28.29% about the total cases of 2440.
- The third affected age by disease is Young Adult (35-50 years) 17.12% about the total case of 1477.
- The fourth affected age by disease is Adult (20-35 years) 14.78% about the total case of 1275.
- The fifth affected age group is Teen (13-19 years) 7.80% about the total case of 673.
- The last affected age group is child (0-12 years) 1.25% about the total case of 108.

4.3.2 Top Ten Disease and Their Cases

- DM (Diabetes Mellitus): 1,126 cases
- HTN (Hypertension): 867 cases
- CVA (Cerebrovascular Accident / Stroke): 681 cases
- AGE (Acute Gastroenteritis): 648 cases
- FEVER: 431 cases
- CKD (Chronic Kidney Disease): 318 cases
- LRTI (Lower Respiratory Tract Infection): 297 cases
- ANEMIA: 237 cases
- BLOOD TRANSFUSION: 227 cases
- AFI (Acute Febrile Illness): 222 cases

4.3.3 Disease Seasonality Analysis

Gastrointestinal Summer spike: Gastrointestinal diseases (AGE) are acutely climate sensitive disease identified during study period. The number of AGE cases more than tripled between summer (364 cases) and autumn/spring (112-116 cases), this shows significant changes in the public health infrastructure during the hot summer season.

Winter Cardiorespiratory Pivot: The season with the lowest patient load, winter (1,694 cases), also has a unique profile of necessary medical intervention. The emergence of LRTI and IHD shows that cold waves transform a baseline number of cases into significant heart and chest related problems.

Static Baseline: Chronic conditions like DM, HTN, and CVA shows an invariant patient load that is distributed equally among the 365 days of the year.

4.4 Forecasting Model Evaluation

The result and the evaluation of time series forecasting model (prophet) used in the study is interpreted given below.

1	Disease	Total_Cases	Train_Months	Test_Months	MAE	RMSE	MAPE(%)
2	POISONING	20	10	3	0.75	0.91	56.94
3	PNEUMONIA	35	10	3	0.94	0.96	37.36
4	POST NATAL	23	10	3	1.07	1.07	85.88
5	CP CHILD	27	10	3	1.07	1.11	50.07
6	TB	76	10	3	1.15	1.36	19.55
7	SEPSIS	57	10	3	1.27	1.37	47.73
8	CONVERSION DISORDER	30	10	3	1.37	1.41	44.4
9	IDA	20	10	3	1.31	1.42	61.62
10	HEADICH	22	7	3	1.3	1.59	107.43
11	EPIGASTRIC PAIN	55	10	3	1.47	1.8	32.75
12	SOB	55	9	3	1.72	1.95	114.48
13	ASTHMA	54	10	3	1.95	1.99	89.89
14	COUGH	27	9	3	1.98	2.08	146.76
15	TIA	57	10	3	1.75	2.16	26.96
16	LRTI	303	10	3	2.12	2.34	10.64
17	FITS	68	10	3	1.87	2.41	25.93
18	DYSENTERY	53	10	3	2.2	2.42	57.5
19	DKA	84	10	3	2.06	2.63	79.13
20	VERTIGO	27	10	3	2.38	2.9	111.78
21	LVF	56	9	3	2.36	2.92	97.18
22	AC PANRITITIS	47	10	3	2.67	2.98	45.66
23	L-MOTION	27	9	3	2.96	3.16	181.3
24	ASPIRATION	40	10	3	2.61	3.25	63.26
25	IHD	218	10	3	3.6	3.7	21.3
26	VOMITING	80	10	3	3.66	3.85	82.72
27	CAD	168	10	3	4.06	4.24	44.85
28	E-FEVER	62	9	3	4.45	4.71	79.2
29	SNAKE BITE	26	9	3	4.49	4.79	157.1
30	OPP	203	10	3	4.67	5.48	31.84
31	HTN	868	10	3	5.33	5.95	8.23
32	COPD	185	10	3	4.94	6.29	53.72
33	MASHROOM POISONING	21	10	3	5.5	6.6	144.74
34	DEPRESSIVE ILLNESS	69	10	3	5.61	6.73	55.15
35	BLOOD TRANSFUSION	228	10	3	7.45	7.87	51.47
36	UTI	185	10	3	9.05	9.48	56.53
37	AFI	222	10	3	9.42	9.91	47.36
38	AKI	62	10	3	9.77	10.48	103.1
39	CVA	686	10	3	14.51	14.67	30.62
40	ANEMIA	237	10	3	15.31	15.85	54.41
41	CKD	319	10	3	15.91	16.2	55.69
42	FEVER	431	10	3	19.62	21.13	82.05
43	DM	1126	10	3	28.37	28.45	30.38
44	AGE	651	10	3	83.1	88.69	107.5
45							

Figure 4.4 Model Evaluation Result

This table shown in fig 4.4 forecasting performance with Prophet for 44 diseases. Evaluation metrics used are MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), MAPE (Mean Absolute Percentage Error). The lower the value the higher the accuracy.

For all the diseases analyzed, Hypertension is with 8.23% MAPE had the lowest errors, followed closely by Lower Respiratory Tract Infection is 10.64%. Both error values are considerably low and represent high confidence and stability of forecasts generated by the

model, suggesting that the pattern of these diseases over time could be very consistently predicted. MAPE for Tuberculosis is 19.55%, Fits is 25.93%, Diabetes Mellitus is 30.38%, Cerebrovascular Accident is 30.62%, Organophosphorus Poisoning is 31.84% and Pneumonia is 37.36% had similarly low to moderate forecasting errors, suggesting a relatively consistent trend and therefore easier and more reliable prediction.

Medium performance predictions of diseases over time were found. MAPE for Sepsis is 47.73%, Blood Transfusion is 51.47%, Chronic Obstructive Pulmonary Disease is 53.72%, Depressive Illness is 55.15%, Poisoning is 56.94% is and Diabetic Ketoacidosis is 79.13%. Though higher than for earlier mentioned diseases, the models were still effective at charting and interpreting trends, even if slightly more errors were being made during each prediction.

Larger forecasting errors were made, MAPE for Headache is 107.43%, Vertigo is 111.78%, Shortness of Breath is 114.48%, Aspiration is 124.31%, Mushroom Poisoning is 144.74%, Cough is 146.76% and Snake Bite is 157.10%. These highly varying predictions can be attributed to significant and rapid fluctuations in the trends and prevalence of these diseases. Possible reasons could include the possibility of outbreak, weather, and seasonality influencing the figures, or simply due to randomness in patient admission of such ailments.

4.4.2.1 Satisfactory Model Result

A model with as small as possible MAE, RMSE, and MAPE is consider to good model.

Explanation:

MAE (Mean Absolute Error) means the forecasting results are very close to the real values.

RMSE (Root Mean Square Error) means big errors in forecast are small.

MAPE (Mean Absolute Percentage Error) means the prediction accuracy in percentage is high.

Based on our results, diseases such as:

HTN (MAPE = 8.23%)

LRTI (MAPE = 10.64%)

TB (MAPE = 19.55%)

could be classified as good predicted diseases due to its relatively small forecasting errors.

The developed forecasting model performs well for several diseases such as HTN, LRTI and TB with relatively low MAE, RMSE and MAPE values. These relatively small error metrics show that the number of predicted diseases matches well with the actual observed ones. It implies that the past dynamics of these diseases are relatively stable and consistent; thus the prediction can be accurate.

4.4.2.2 Inadequate Model Result

When MAE, RMSE and MAPE values are high, a forecasting model can be classified as poor.

Reason

The high value for MAE implies large discrepancies between predicted and actual values.

The high value for RMSE implies substantial amount of forecasting errors being present.

The high value for MAPE implies a considerable distance between predictions and actual observations.

Our results

The following diseases have shown poor forecasting performance:

L-MOTION (MAPE = 181.30%)

Snake Bite (MAPE = 157.10%)

Cough (MAPE = 146.76%)

Mushroom Poisoning (MAPE = 144.74%)

The forecasting model showed poor forecasting performance in predicting some diseases, namely L-MOTION, Snake Bite, Cough and Mushroom Poisoning with high MAE, RMSE and MAPE values. These error levels represent significant gaps between the predicted and observed disease cases. This could be due to the irregular nature of disease occurrences, abrupt change in the number of admissions or insufficient history of observation or the presence of outlier, which makes it difficult for the model to detect a consistent trend and generate reliable forecasts.

4.4.2.3 Why Some Diseases Have High MAPE Despite Low MAE

The following example shows for diseases with very few monthly cases, even a small prediction error can produce a very high MAPE.

Example:

- Actual cases = 1
- Predicted cases = 3
- Error = 2 cases only

But:

$$\text{MAPE} = \frac{|1-3|}{1} \times 100 = 200\%$$

Therefore, diseases with low case counts often show extremely high MAPE values even when the absolute error is small.

4.4.2.4 Interpretation of Some Forecasting Figures

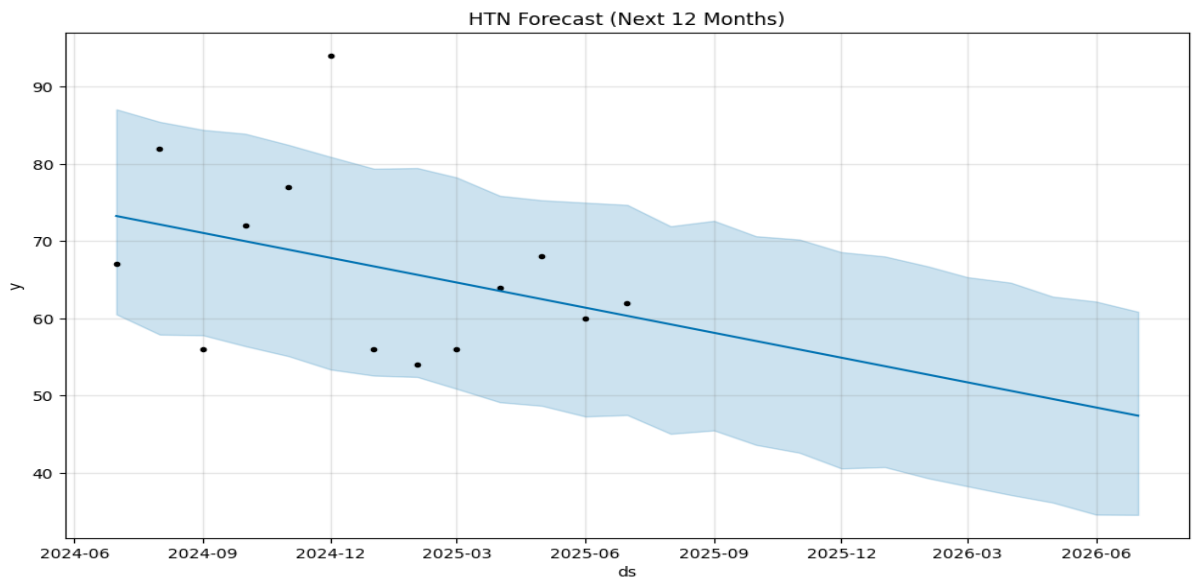


Figure 4.5 HTN Forecast

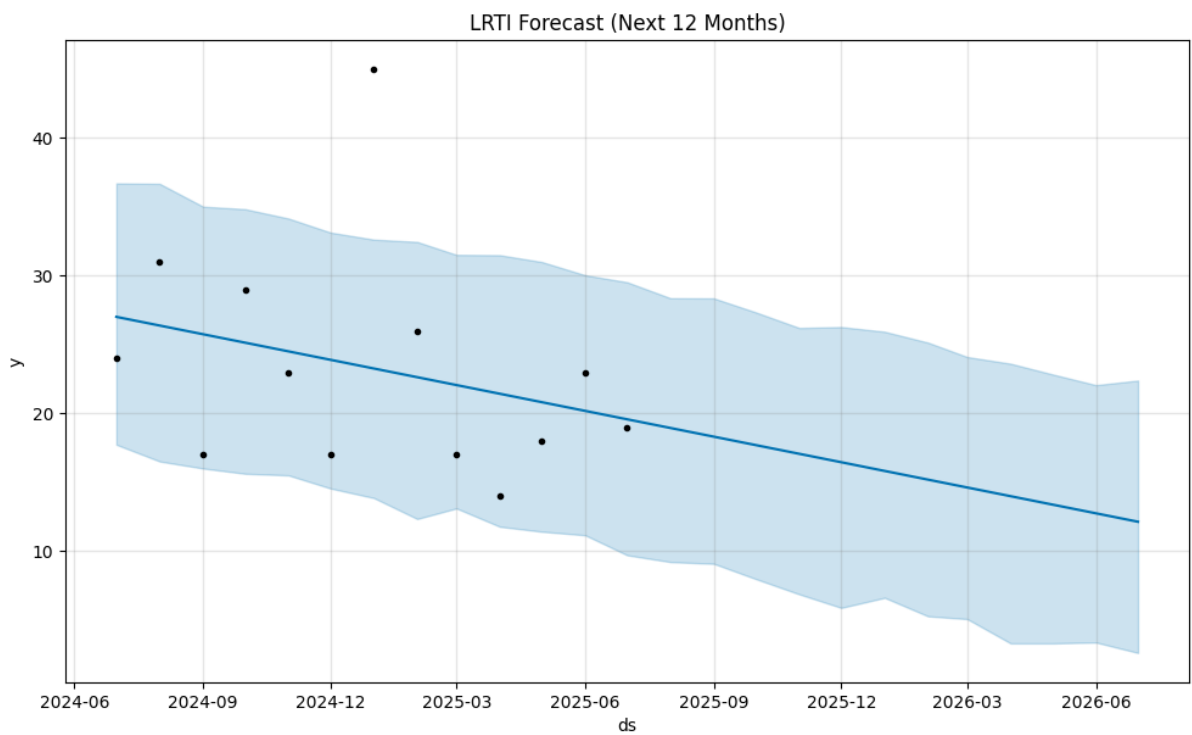


Figure 4.6 LRTI Forecast

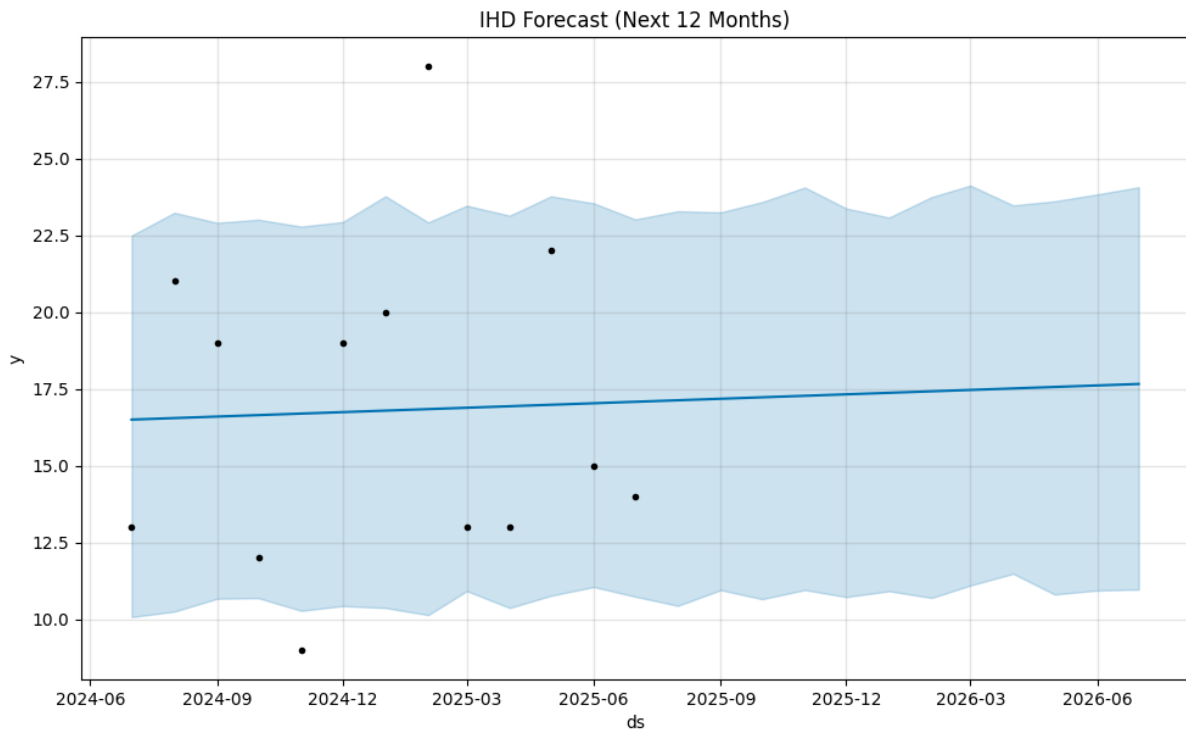


Figure 4.7 IHD Forecast

The given figures shown in 4.5, 4.6 and 4.7 shows the forecasting of HTN, LRTI and TB for next 12 months respectively.

Key Components

The key components of the forecasting figure are given below:

1. X-axis (ds)

The datestamp ds, here we are monitoring these trends. On these particular plots, this takes the month through mid-2024 and into mid-2025, then continues for the next 12 months in future predictions until mid-2026.

2. Y-axis (y)

The Y-axis is the independent variable, y, representing the numerical data that the model is attempting to track and predict. This represents the number of patient admissions or records of disease in the department of medicine, based on the exact parameters of our data.

3. Black Dots Data points

The black dots represent the recorded data falls and is plotted, against where the predictive model estimates this data would lie.

4. Dark Blue Line The trend line ($\hat{Y}$)

The Dark Blue Line shows the actual predicted data of how we think future medical admissions will run based on previous data, where we remove all the "randomness" and season fluctuations of real life data to give an estimated central prediction.

5. Light Blue area Uncertainty Interval

It represents the confidence level of the model in prediction. This interval represents how the predicted line would differ if the variability in past trends continue and if the changeability in the rate of change is recorded to simulate variations into the future. It is also useful for discussing what level of uncertainty we have within the trends, if you can trust these predictions, and how unpredictable patterns are likely to become. As the light blue area continues to spread outwards, this demonstrates clearly that the further out the model is predicting, the less certainty it has in what will happen with your data.

1.Hypertension (HTN)

Figure (4.5) presents the forecasted trend of HTN cases for the next 12 months using the Prophet forecasting model.

The forecast represents a moderate decrease in HTN case numbers during the forecasted time frame. Examining monthly values historically, there were highs around 94 cases, lows around 54 cases, and averages throughout. Based on the model, the forecast predicts case numbers decreasing from roughly 73 at the beginning of the forecast time frame down to about 47 at the end of the forecast time frame.

The confidence interval is fairly wide throughout the entire forecasted period. This uncertainty is caused mostly by the relatively small number of historical cases that the model is trained on, as well as the fluctuations among those cases. Despite that, the trend of the forecast itself indicates that a slight decrease in the total cases in the Medical Department might be expected. This could help to plan resource distribution, manning, and case management for the coming months. These numbers must be used carefully as the relatively short duration of this dataset may prove it to have poor long term predictive value.

Key Finding

- Forecast trend: Decreasing
- Starting forecast value: ≈ 73 cases
- Ending forecast value: ≈ 47 cases
- Overall change: Decrease of approximately 26 cases
- Confidence interval: Wide, indicating moderate forecasting uncertainty
- Implication: Expected decline in HTN patient admissions over the next 12 months, subject to future healthcare and environmental conditions.

2.Lower Respiratory Tract Infection (LRTI)

According to fig (4.6), the projection predicts an ongoing negative trend of LRTI occurrences for the next 12 months.

The observed time-series starting values are: it starts off with a forecast $Y = 27$ cases in Jun 2024 and is then estimated to decrease steadily to approximately $Y = 12$ cases by July 2026. This negative trend can be interpreted as a depleting susceptible pool, long-term effect of preventive healthcare activities, or a broader change in the surrounding/seasonality reducing the number of transmissions.

The observed points (shown as black dots) do show some level of variation; a peak is observed around Jan 2025 where an observation reaches a value close to 45, far outside of the upper bound of the model uncertainty interval, probably due to a season-related episode or unmodeled factor (weather, local occurrence).

The shaded blue area represents the 95% uncertainty interval. The model's prediction shows roughly equal width through time, but with an increasing width towards the latter part of the prediction period (mid-2026), due to compounding error in longer-term predictions.

Key Finding

- Forecast trend: Decreasing
- Starting forecast value: ≈ 27 cases
- Ending forecast value: ≈ 45 cases
- Overall change: Decrease of approximately 12 cases
- Confidence interval: Wide, indicating moderate forecasting uncertainty
- Implication: Expected decline in LRTI patient admissions over the next 12 months, subject to future healthcare and environmental conditions.

Ischemic Heart Disease (IHD)

The figure (4.7) shows a 12-month forward projection for Ischemic Heart Disease (IHD) cases. The analysis evaluates historical data patterns to predict patient volume trends from mid-2025 through mid-2026. Based on the IHD interpretation you provided earlier, the paragraph in the same style would be:

As illustrated in the IHD forecast plot, the predictive model projects a relatively stable but slightly increasing trend in Ischemic Heart Disease (IHD) cases over the next 12 months. The forecast begins with an expected value ($\hat{Y}$) of approximately 17 cases and gradually increases to nearly 18 cases by July 2026. This mild upward trend suggests that the burden of IHD is expected to remain largely stable, with only marginal growth in patient admissions over the forecast period. The historical observations (represented by black dots) exhibit noticeable variability, with a peak reaching approximately 28 cases around January/February 2025 and a minimum of about 9 cases around October 2024. These fluctuations may be associated with seasonal influences, demographic factors, lifestyle-related cardiovascular risks, or other external factors not explicitly captured by the model. The shaded blue region represents the 95% uncertainty interval, which remains relatively consistent throughout the forecasting horizon. The stable width of the confidence interval indicates moderate forecasting uncertainty and suggests that future case variability is expected to remain similar to historical patterns. Overall, the forecast indicates that healthcare facilities should maintain sufficient cardiovascular care resources to accommodate a baseline of approximately 15–18 IHD cases per month while remaining prepared for occasional increases approaching the upper confidence limits.

Key Finding

- Forecast trend: stable, marginally upward linear trend
- Starting forecast value: ≈ 17 cases
- Ending forecast value: ≈ 18 cases
- Overall change: Increase of 1 cases

- Confidence interval: Wide, indicating moderate forecasting uncertainty
- Implication: Expected increase in IHD patient admissions over the next 12 months, subject to future healthcare and environmental conditions.

4.4.2.5 Effect of Limited Historical Data on Forecasting Performance

For the forecasting model development, 13 months of historical diseases records were used and nearly 10 months were used for training and 3 months for testing. The model was able to find overall disease trends but limited length of data perhaps lowered down prediction for some diseases. Time-series forecasting models perform best with longer histories as they can discover the long-term patterns, seasonal variations etc.

The usual prediction practice in forecasting is recommended with at least 24-36 months of past data for reliable monthly prediction when seasonality is expected. The small data history length of 13 months restricted the models to discover the disease behaviors in the long term, which may increase the forecasting error for less consistent diseases. Predictability was highest for regular disease behaviors, whereas prediction performance was lower for infrequent diseases.

Therefore, prediction results need to be assessed considering the small history length and higher prediction performance could be achieved with multiple years of disease records.

Overall Conclusion of Model performance

The forecasting model achieved satisfactory performance for diseases with stable and consistent occurrence patterns, while prediction accuracy decreased for diseases characterized by low frequencies, irregular fluctuations or sudden outbreaks.

Consequently, the model is more reliable for forecasting common and regularly occurring diseases than for rare or highly variable disease conditions.

4.6 Dashboard Design Interpretation



Figure 4.8 Dashboard

This dashboard can offer the overall views of patient flow, disease, treatment progress and disease forecasting in the Medical Department. As a decision-support tool, it gives us valuable medical information by interactive visualizations and summary statistics.

In the top part of the dashboard there are some important performance indicators which show an overview about the operation status of hospital. They can be an overview about the total patient number, number of admissions, number of discharges, success of treatment, mortality and average length of stay. Users can have a general understanding of the current department state in this part without browsing detailed record.

This is a patient flow chart showing the trend of admission during certain periods of time. Users can see patient coming in to hospital trends in this visualization and know the increase or decrease of department work load during different times. Additional chart about patient age groups and genders shows characteristics of patients in terms of demographics, and users can be aware of the demographic groups mostly served by health care resources. The disease trend section focuses on analyzing disease prevalence and patterns. The top diseases chart is intended to highlight the most frequently reported diseases, while the disease trend chart displays how disease occurrences change over time. These two types of charts are used to find out most important health problems, and also track down the disease performance on study patients, and then show by a disease share chart. It shows how much contribution each disease to disease load.

The treatment result part is designed to measure how well the medical care given to patients. By the visualizations, users can find out the outcome of the patients, and compare the treatments of the disease, hence the results can be used to review healthcare performance.

The forecasting part takes the step further by showing the prediction about the disease trend in the future. They aim to show the disease performance predicted on a future date, and also the prediction of disease occurrences over a range. Hospital managers can use this information to plan their resources for the future healthcare needs.

The filter functions on the dashboard include time range, age, sex, and specific disease. Users can use it to investigate what they want, and make some analysis according to their own needs.

The dashboard's main purpose is to summarize various detailed hospital data, in the form of visualizations and present it in an intuitive interface to users for analyzing patient flow, disease trends, treatment performance, future healthcare prediction, etc.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter concludes the study with a brief outline of the research that was conducted, conclusions and recommendations. Future research ideas are also proposed in this section. The study focused on analyzing patient flow patterns, disease trends, treatment outcomes, disease forecasting and dashboard development using historical patient records from the Medical Department of DHQ Hospital Timergara.

5.2 Summary of the Study

This research aimed to examine historical patient data so that trends in patient flow and disease pattern can be observed and utilized to improve care within Medical Department. The stages undertaken to achieve this aim were the collection, pre-processing, cleaning, analyzing, visualizing, forecasting and dashboard design stages. The stages undertaken included examining patient flow and demographic information of patients, trends of the diseases in the Medical Department, the treatments outcomes within Medical Department and forecasting the occurrence of diseases in the future. A conceptual framework of dashboard was developed and the potential use of this could also be analyzed.

5.3 Conclusion

The project successfully explained the application of data analytics for evaluating the trend of patients flow and diseases within Medical Department at DHQ Hospital Timergara. It gives us better insight into the patient's arrival trends, age distribution of patients, diseases that are more prevalent and their treatment outcomes. Disease trends illustrated what are the dominant issues faced by the patients and prediction forecasts further expand our observation by estimating the occurrences of diseases.

The dashboard is useful in simplifying large set of healthcare data in the form of attractive and interactive display that can help in decision making among healthcare providers and management. It integrates the information of patient flow trends, diseases, and treatment outcomes and prediction into a single and coherent dashboard.

Overall, the study highlights the importance of utilizing historical healthcare data to generate meaningful insights that contribute to improved healthcare management, operational efficiency and strategic planning.

5.4 Recommendations

The below recommendation are proposed on the basis of this study:

1. Continuous Monitoring of Disease Trends

The hospital administration should keep a watch on the diseases trend so as to tackle with the health challenges that will arise in future.

2. Data-Driven Decision Making

Data should support health care planning and resource allocation decisions instead of reliance on tradition methods.

3. Improved Data Management

Consistent and accurate data entry practices should be managed to improve the reliability and quality of data for future analysis.

4. Implementation of Interactive Dashboards

This dashboard that we proposed should be built to let the healthcare administrators obtain the critical information on patients and disease.

5. Resource Planning Based on Forecasting

The forecast results are expected to provide a basis to future planning on medical staff, drugs, beds and others healthcare resources.

6. Focus on High-Burden Diseases

Special attention should be given to diseases that consistently contribute to a large proportion of hospital admissions, including preventive measures and awareness programs.

5.5 Future Work

Although this study achieved its objectives and several opportunities exist for future improvement and extension.

1. Real-Time Dashboard Development

Future work could involve the development of a real-time dashboard linked directly to hospital databases to monitor activity continuously.

2. Advanced Forecasting Models

More machine learning and deep learning techniques can also be experimented with for better prediction.

3. Inclusion of Additional Departments

It would be interesting in future research to be able to incorporate results from other hospital departments and be able to look at overall hospital trends.

4. Integration with Hospital Information Systems

The dashboard can also be incorporated into the hospital management systems to allow for automated reporting and analysis.

5. Predictive Risk Analysis

Research in the future might even take it a further step in predicting patient risk levels, disease severity and outcomes of health care with advanced predictive analytic methodologies.

6. Geographic Disease Analysis

The use of spatial and geographical analysis may also be included in an attempt to identify clusters of disease and provide support for public health intervention measures.

5.6 Final Remarks

From this study, we came to know that Healthcare Data Analytics is valuable in understanding the flow of patients and disease pattern in Medical department of DHQ Hospital Timergara and the result of analysis, prediction and dashboard can be used for betterment of the healthcare management and the decision making can be based on evidence. The model given herein can provide platform for future data analytics work in healthcare and the use of hospital data.

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